

Problem Set 1 Veronica Gebbia

★ pg 814 exercise 1: write out the following summations.

$$a) \sum_{k=1}^3 k^2 \rightarrow 1^2 + 2^2 + 3^2 = 14$$

$$b) \sum_{k=1}^3 a k^2 \rightarrow a_1^2 + a_2^2 + a_3^2$$

$$c) \sum_{j=1}^3 \sum_{k=1}^3 a_{jk} b_{kj} \rightarrow \sum_{j=1}^3 a_j (b_{1j} + b_{2j} + b_{3j}) \rightarrow$$

$$= (a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}) + (a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}) + (a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33})$$

$$d) \left(\sum_{j=1}^3 a_j \right)^2 = \sum_{j=1}^3 \sum_{k=1}^3 a_j a_k \rightarrow a_1^2 + a_1 a_2 + a_1 a_3 + a_2 a_1 + a_2^2 + a_2 a_3 + a_3 a_1 + a_3 a_2 + a_3^2$$

★ pg 814 exercise 2: $v_x = 1, v_y = 2, v_z = -5$

$$w_x = 3, w_y = -1, w_z = 1$$

$$a) (v \cdot w) \rightarrow (1 \cdot 3) + (2 \cdot -1) + (-5 \cdot 1) = -4$$

$$b) [v \times w] \rightarrow \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 2 & -5 \\ 3 & -1 & 1 \end{vmatrix}$$

$$\rightarrow (2 \cdot 1 - (-5 \cdot -1))\hat{x} - (1 \cdot 1 - (-5 \cdot 3))\hat{y} + (3 \cdot -1 - (-2 \cdot 3))\hat{z} = -3\hat{x} - 16\hat{y} - 7\hat{z}$$

$$c) \text{length of } v = |v|$$

$$|v| = \sqrt{(1)^2 + (2)^2 + (-5)^2} = \sqrt{30}$$

$$d) (\hat{f}_x \cdot v) \quad \hat{f}_x = (1, 0, 0)$$

$$\hat{f}_x \cdot v \rightarrow (1 \cdot 1) + (0 \cdot 2) + (0 \cdot -5) = 1$$

$$e) [f_x \times w] \rightarrow f_x = (1, 0, 0) \quad w = (3, -1, 1)$$

$$\begin{array}{ccc} f_x & f_y & f_z \\ 1 & 0 & 0 \end{array} \rightarrow \begin{array}{ccc} & & -1-0= \\ f_x(0 \cdot 1 - (0 \cdot -1)) - f_y(1 \cdot 1 - (0 \cdot 3)) + f_z(1 \cdot -1 - (0 \cdot 3)) \\ 3 & -1 & 1 \end{array} \quad \Delta f_x - f_y - f_z \rightarrow -f_y - f_z / (0, -1, -1)$$

$$f) \phi_{vw} \rightarrow \cos \phi_{vw} = \frac{v \cdot w}{|v||w|} \rightarrow -4$$

$$|v||w| \rightarrow \sqrt{30}|w|$$

$$|w| = \sqrt{3^2 + (-1)^2 + 1^2} = \sqrt{11}$$

$$\cos \phi_{vw} = -4 / \sqrt{30 \cdot 11} \rightarrow -4 / \sqrt{330} \rightarrow$$

$$\phi_{vw} = \cos^{-1}(-4 / \sqrt{330}) \rightarrow 102.72^\circ - 180^\circ = 77.28^\circ = \phi_{vw}$$

$$g) [r \times v] \quad r = \text{position} \quad r = (x, y, z)$$

$$\begin{array}{ccc} f_x & f_y & f_z \\ x & y & z \end{array} \rightarrow \begin{array}{ccc} f_x(y \cdot -5 - (z \cdot 2)) - f_y(x \cdot -5 - (z \cdot 1)) + f_z(x \cdot 2 - (y \cdot 1)) \\ 1 & 2 & -5 \end{array} \quad \begin{array}{ccc} f_x(-5y - 2z) - f_y(-5x - z) + f_z(2x - y) \\ (-5y - 2z)f_x + (5x + z)f_y + (2x - y)f_z / (-5y - 2z, 5x + z, 2x - y) \end{array}$$

★ pg 819 exercise 1

$$\tau \text{ comps: } \tau_{xx} = 3 \quad \tau_{xy} = 2 \quad \tau_{xz} = -1$$

$$\tau_{yx} = 2 \quad \tau_{yy} = 2 \quad \tau_{yz} = 1$$

$$\tau_{zx} = -1 \quad \tau_{zy} = 1 \quad \tau_{zz} = 4$$

$$v \text{ comp: } v_x = 5 \quad v_y = 3 \quad v_z = -2$$

$$a) [\tau \cdot v] \quad \begin{bmatrix} (3 \cdot 5) + (2 \cdot 3) + (-1 \cdot -2) \\ (2 \cdot 5) + (2 \cdot 3) + (1 \cdot -2) \\ (-1 \cdot 5) + (1 \cdot 3) + (4 \cdot -2) \end{bmatrix} = \begin{bmatrix} 23 \\ 14 \\ -10 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 3 & 2 & -1 & 5 \\ 2 & 2 & 1 & 3 \\ -1 & 1 & 4 & -2 \end{array} \right] \nearrow$$

$$b) [\psi \cdot \chi] \Rightarrow \langle \psi | \chi \rangle = [5 \ 3 \ -2] \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \text{ symmetrical w/ part a} \\ \text{but makes } [23 \ 14 \ -10]$$

$$c) (\chi : \chi) \Rightarrow \text{tr}(\chi^T \chi) \Rightarrow \\ (3 \cdot 3) + (2 \cdot 2) + (-1 \cdot -1) + (2 \cdot 2) + (2 \cdot 2) + (1 \cdot 1) + (-1 \cdot -1) + \\ (1 \cdot 1) + (4 \cdot 4) = 41$$

$$d) (\psi \cdot [\chi \cdot \psi]) \Rightarrow \langle \psi | \chi | \psi \rangle$$

$$\begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 23 \\ 14 \\ -10 \end{bmatrix} \rightarrow (5 \cdot 23) + (3 \cdot 14) + (-2 \cdot -10) = 177$$

$$e) \psi \psi \Rightarrow |\psi\rangle\langle\psi| \Rightarrow \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} [5 \ 3 \ -2] \Rightarrow$$

$$\Rightarrow \begin{bmatrix} (5 \cdot 5) & (5 \cdot 3) & (5 \cdot -2) \\ (3 \cdot 5) & (3 \cdot 3) & (3 \cdot -2) \\ (-2 \cdot 5) & (-2 \cdot 3) & (-2 \cdot -2) \end{bmatrix} \Rightarrow \begin{bmatrix} 25 & 15 & -10 \\ 15 & 9 & -6 \\ -10 & -6 & 4 \end{bmatrix}$$

$$f) [\chi \cdot \delta_x] \Rightarrow |\chi| \delta_x \Rightarrow \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} = \leftarrow$$

★ pg 819 exercise 2

$$a) [\langle f_1 | f_2 \cdot f_2 \rangle \times f_1] \Rightarrow (|a\rangle \langle b|) |w\rangle = |a\rangle \langle b| w\rangle$$

$$(|f_1\rangle \langle f_2|) |f_2\rangle = |f_1\rangle \underbrace{\langle f_2 | f_2 \rangle}_1 = |f_1\rangle \rightarrow$$

$$|f_1\rangle \times |f_1\rangle = |0\rangle \text{ vector}$$

$$b) (f: f_1 f_2) \Rightarrow \text{tr}(|f_1\rangle \langle f_2|) = \langle f_2 | f_1 \rangle = 0 \text{ scalar}$$

$$c) (f:f) \Rightarrow \text{tr}(I) = 3$$

$$d) \{ f \cdot f \} \Rightarrow I \cdot I = I = |f|$$

★ pg 823 exercise 3:

$$(\nabla \cdot v) \hat{=} \nabla v \hat{=} [\nabla \cdot v v]$$

$$\text{Cartesian coord: } \langle \text{div } v \rangle = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$|\nabla v| = \frac{\partial v_i}{\partial x_j} \quad \text{div} |v\rangle \langle v| = |\nabla v| |v\rangle + \langle \text{div } v \rangle |v\rangle$$

$\hookrightarrow \nabla v \qquad \qquad \qquad \hookrightarrow [\nabla \cdot v v]$

$$a) v \text{ for } (b_y, 0, 0)$$

$$\langle \text{div } v \rangle = 0$$

$$|\nabla v| = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow = b |f_x\rangle \langle f_y|$$

$$\text{div}(|v\rangle \langle v|) = \langle 0, 0, 0 \rangle = |0\rangle$$

$$b) v \text{ for } (b_x, 0, 0)$$

$$\langle \text{div } v \rangle \Rightarrow \frac{\partial v_x}{\partial x} = b = v_x = b + 0 + 0$$

$$\frac{\partial v_y}{\partial y} = 0 = v_y$$

$$\frac{\partial v_z}{\partial z} = 0 = v_z \quad \langle \text{div } v \rangle = b$$

$$|\nabla v| = \begin{bmatrix} b & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = b|dx\rangle\langle dy|$$

$$\text{div}(|v\rangle\langle v|) = \langle 2b^2 x, 0, 0 \rangle = 2b^2 x |dx\rangle$$

c) v for $(by, bx, 0)$

$$\langle \text{div } v \rangle \Rightarrow \frac{\partial v_x}{\partial x} = 0 \quad = 0 = \langle \text{div } v \rangle$$

$$\frac{\partial v_y}{\partial y} = 0$$

$$\frac{\partial v_z}{\partial z} = 0$$

$$|\nabla v| = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = b|dx\rangle\langle dy| + b|dy\rangle\langle dx|$$

$$\text{div}(|v\rangle\langle v|) = 0 \cdot |v\rangle = 0 \Rightarrow \langle b^2 x, b^2 y, 0 \rangle$$

$$\text{div}(|v\rangle\langle v|) = b^2 x |dx\rangle + b^2 y |dy\rangle \quad \leftarrow \begin{matrix} x \\ y \\ z \end{matrix}$$

d) v for $v_x = -by$ $v_y = bx$ $v_z = 0$

$$\langle \text{div } v \rangle \Rightarrow \frac{\partial v_x}{\partial x} = 0; \frac{\partial v_y}{\partial y} = 0; \frac{\partial v_z}{\partial z} = 0$$

$$\langle \text{div } v \rangle = 0$$

$$|\nabla v| = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -b|dx\rangle\langle dy| + b|dy\rangle\langle dx| = |\nabla v|$$

$$\text{div}(|v\rangle\langle v|) \Rightarrow \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -by \\ bx \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} -b^2 x \\ -b^2 y \\ 0 \end{bmatrix} \Rightarrow \langle -b^2 x, -b^2 y, 0 \rangle$$

$$\text{div}(|v\rangle\langle v|) = -b^2 x |dx\rangle - b^2 y |dy\rangle$$

★ pg 823 exercise 9:

a) $\nabla 1/r = -r/r^3 \rightarrow g(r, \theta, \phi) = 1/r$

$$\partial/\partial r (1/r) = -1/r^2, \partial/\partial \theta (1/r) = 0, \partial/\partial \phi (1/r) = 0$$

so $\langle \nabla 1/r | = -1/r^2 \langle f_r |$ and $\langle r | = r \langle f_r | \rightarrow \langle f_r | = 1/r \langle r |$

$$\langle \nabla 1/r | = -1/r^3 \langle r | \rightarrow \nabla (1/r) = -r/r^3$$

b) $\nabla f(r) = 1/r df/dr \rightarrow \langle \nabla f(r) | = \rightarrow$

$$\partial f/\partial r \langle f_r | + \underbrace{1/r \partial f/\partial \theta \langle f_\theta |}_{0} + \underbrace{1/r \sin \theta \partial f/\partial \phi \langle f_\phi |}_{0}$$

$$\langle \nabla f(r) | = df/dr \langle f_r | \rightarrow \langle f_r | \text{ to } 1/r \langle r | \rightarrow$$

$$\langle \nabla f(r) | = 1/r df/dr \langle r | \text{ vector} = |\nabla f(r)\rangle = 1/r df/dr |r\rangle$$

c) $\nabla(a \cdot r) = a$ $a = \text{const. vector}$

$$(a \cdot r) = \langle a | r \rangle \rightarrow \langle a | \nabla r | \rightarrow \nabla r = |f| \rightarrow$$

$$\langle a | \nabla r | = \langle a | f | = \langle a |$$

$$\langle \nabla(a \cdot r) \rangle = |a\rangle$$

★ pg 823 exercise 10:

a) $\partial/\partial t \rho v = -[\nabla \cdot \rho v v] - \nabla p - [\nabla \cdot \tau] + \rho g$

$$\frac{\partial \rho |v\rangle}{\partial t} = -|\text{div} \rho |v\rangle \langle v| - |\nabla p\rangle - |\text{div} \tau| + \rho |g\rangle$$

1) $\langle f_x | \partial/\partial t \rho |v\rangle = \partial/\partial t \rho \langle f_x | v \rangle = \partial(\rho v_x)/\partial t$

2) $\langle f_x | \text{div} [\rho |v\rangle \langle v|] = \sum_{i=1}^3 \frac{\partial \rho \langle f_x | v \rangle \langle v | f_i \rangle}{\partial q_i} \rightarrow$

$$= \frac{\partial(\rho v_x v_x)}{\partial x} + \frac{\partial(\rho v_x v_y)}{\partial y} + \frac{\partial(\rho v_x v_z)}{\partial z}$$

3) $\langle f_x | \nabla p \rangle = \partial p / \partial x$

$$4) \langle \delta x | \text{div} | \zeta | \rangle = \sum_{i=1}^3 \frac{2 \langle \delta x | \zeta | \delta i \rangle}{2q_i} \rightarrow \frac{\partial \zeta_{xx}}{\partial x} + \frac{\partial \zeta_{xy}}{\partial y} + \frac{\partial \zeta_{xz}}{\partial z}$$

becomes :

$$\frac{\partial(\rho v_x)}{\partial t} = - \left[\frac{\partial(\rho v_x v_x)}{\partial x} + \frac{\partial(\rho v_x v_y)}{\partial y} + \frac{\partial(\rho v_x v_z)}{\partial z} \right] \rightarrow$$

x-comp $- \frac{\partial \rho}{\partial x} - \left(\frac{\partial \zeta_{xx}}{\partial x} + \frac{\partial \zeta_{xy}}{\partial y} + \frac{\partial \zeta_{xz}}{\partial z} \right) + \rho g_x$

* same steps for y and z and they become $\rightarrow$

$$\frac{\partial(\rho v_y)}{\partial t} = - \left[\frac{\partial(\rho v_y v_x)}{\partial x} + \frac{\partial(\rho v_y v_y)}{\partial y} + \frac{\partial(\rho v_y v_z)}{\partial z} \right] \rightarrow$$

$$- \frac{\partial \rho}{\partial y} - \left(\frac{\partial \zeta_{yx}}{\partial x} + \frac{\partial \zeta_{yy}}{\partial y} + \frac{\partial \zeta_{yz}}{\partial z} \right) + \rho g_y$$

$$\frac{\partial(\rho v_z)}{\partial t} = - \left[\frac{\partial(\rho v_z v_x)}{\partial x} + \frac{\partial(\rho v_z v_y)}{\partial y} + \frac{\partial(\rho v_z v_z)}{\partial z} \right] \rightarrow$$

$$- \frac{\partial \rho}{\partial z} - \left(\frac{\partial \zeta_{zx}}{\partial x} + \frac{\partial \zeta_{zy}}{\partial y} + \frac{\partial \zeta_{zz}}{\partial z} \right) + \rho g_z$$

$$b) \zeta = -\mu \nabla^2 \mathbf{v} + (\nabla \mathbf{v})^+ - 2/3 (\nabla \cdot \mathbf{v}) \delta \rightarrow$$

$$|\zeta| = -\mu (|\nabla \mathbf{v}| + |\nabla \mathbf{v}|^+ + (2/3\mu - k) \langle \text{div} \mathbf{v} \rangle |\delta|) \rightarrow$$

in cartesian coord. $\rightarrow$

$$\begin{bmatrix} \zeta_{xx} & \zeta_{xy} & \zeta_{xz} \\ \zeta_{yx} & \zeta_{yy} & \zeta_{yz} \\ \zeta_{zx} & \zeta_{zy} & \zeta_{zz} \end{bmatrix} = -\mu \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} + \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

$\rightarrow + (2/3\mu - k) \rightarrow$ last part, next page $\rightarrow$

$$\langle \text{div } v \rangle |f| \rightarrow \begin{bmatrix} \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) & 0 & 0 \\ 0 & \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) & 0 \\ 0 & 0 & \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \end{bmatrix}$$

$$\tau_{xx} = -2\mu \left(\frac{\partial v_x}{\partial x} \right) + \left(\frac{2}{3}\mu - k \right) \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$\tau_{yy} = -2\mu \left(\frac{\partial v_y}{\partial y} \right) + \left(\frac{2}{3}\mu - k \right) \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$\tau_{zz} = -2\mu \left(\frac{\partial v_z}{\partial z} \right) + \left(\frac{2}{3}\mu - k \right) \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$\tau_{xy} = -\mu \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right) = \tau_{yx}$$

$$\tau_{xz} = -\mu \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) = \tau_{zx}$$

$$\tau_{yz} = -\mu \left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right) = \tau_{zy}$$

★ pg 825 exercise 1

$$v = f_1 x_1 + f_2 x_3 + f_3 x_2$$

$$x_1 = 0, x_1 = 1, x_2 = 0, x_2 = 2, x_3 = 0, x_3 = 4$$

$$\int_S \langle v | n \rangle dS = \int_V \langle \text{div } v \rangle dV$$

$$\int_S \langle v | n \rangle dS = - \int_{x_1=0}^1 \int_{x_2=0}^2 \int_{x_3=0}^4 dx_2 dx_3 v_1 + \int_{x_1=1}^1 \int_{x_2=0}^2 \int_{x_3=0}^4 dx_2 dx_3 v_1 \rightarrow$$

$$- \int_{x_2=0}^2 \int_{x_3=0}^4 dx_1 dx_3 v_2 + \int_{x_2=2}^2 \int_{x_3=0}^4 dx_1 dx_3 v_2 - \int_{x_3=0}^4 \int_{x_2=0}^2 dx_1 dx_2 v_3 \rightarrow$$

$$+\iint_{x_3=4} dx_1 dx_2 v_3 \rightarrow -\underbrace{\int_0^2 \int_0^4 dx_2 dx_3 x_1}_{=0} \rightarrow -\int_0^2 \int_0^4 dx_2 dx_3 (0)$$

$$\int_0^2 \int_0^4 dx_2 dx_3 x_1 \rightarrow \int_0^2 \int_0^4 dx_2 dx_3 (1) \rightarrow (2 \cdot 4 \cdot 1) = 8$$

$$-\int_0^1 \int_0^4 dx_1 dx_3 x_3 \rightarrow -1 \left[\frac{1}{2} x_3^2 \right]_0^4 = -8$$

$$\int_0^1 \int_0^4 dx_1 dx_3 x_3 \rightarrow 1 \left[\frac{1}{2} x_3^2 \right]_0^4 = 8$$

$$-\int_0^1 \int_0^2 dx_2 dx_1 x_2 \rightarrow -1 \left[\frac{1}{2} x_2^2 \right]_0^2 = -2$$

$$\int_0^1 \int_0^2 dx_2 dx_1 x_2 \rightarrow 1 \left[\frac{1}{2} x_2^2 \right]_0^2 \rightarrow 2$$

$$0 + 8 - 8 + 8 - 2 + 2 = 8$$

$$\langle \text{div } v \rangle = \sum_{i=1}^3 \frac{\partial v_i}{\partial x_i} = \frac{\partial x_1}{\partial x_1} + \frac{\partial x_3}{\partial x_3} + \frac{\partial x_2}{\partial x_3} = 1 + 0 + 0$$

$\downarrow$
 $=1$

$$\int_V dv \langle \text{div } v \rangle = \int_0^1 \int_0^2 \int_0^4 (dx_3 dx_2 dx_1) 1 \Rightarrow (1 \cdot 2 \cdot 4)$$

$$\rightarrow = 8 \quad 8 = 8 \checkmark \quad \int_S dS \langle v | n \rangle = \int_V dv \langle \text{div } v \rangle = 8$$

★ pg 839 exercise 1:

$$\mathbf{v} = d/dt \mathbf{r} = \hat{r} \dot{r} + r \hat{\theta} \dot{\theta} + \hat{z} \dot{z}$$

$$|\mathbf{r}\rangle = r |\hat{r}\rangle + z |\hat{z}\rangle$$

$$|v\rangle = \frac{d|\mathbf{r}\rangle}{dt} = \frac{dr}{dt} |\hat{r}\rangle + r \left(\frac{d|\hat{r}\rangle}{dt} \right) + \frac{dz}{dt} |\hat{z}\rangle$$

cylindrical coord: $\frac{\partial |\hat{r}\rangle}{\partial \theta} = |\hat{\theta}\rangle$ $\frac{\partial |\hat{\theta}\rangle}{\partial \theta} = -|\hat{r}\rangle$

$$\frac{\partial |\hat{z}\rangle}{\partial \theta} = 0 \quad |\mathbf{r}\rangle = r |\hat{r}\rangle + z |\hat{z}\rangle$$

$$\frac{d|\hat{r}\rangle}{dt} = \dot{\theta} |\hat{\theta}\rangle \quad \frac{d|\hat{\theta}\rangle}{dt} = -\dot{\theta} |\hat{r}\rangle \quad \frac{d|\hat{z}\rangle}{dt} = 0$$

$$|v\rangle = dr/dt |\hat{r}\rangle + r (d|\hat{r}\rangle/dt) + (dz/dt) |\hat{z}\rangle \\ = \dot{r} |\hat{r}\rangle + r (\dot{\theta} |\hat{\theta}\rangle) + \dot{z} |\hat{z}\rangle$$

$$|v\rangle = \dot{r} |\hat{r}\rangle + r \dot{\theta} |\hat{\theta}\rangle + \dot{z} |\hat{z}\rangle$$

next:

$$\mathbf{a} = \hat{r}(\ddot{r} - r\dot{\theta}^2) + \hat{\theta}(r\ddot{\theta} + 2\dot{r}\dot{\theta}) + \hat{z}\ddot{z}$$

$$|a\rangle = d|v\rangle/dt$$

$$d/dt (\dot{r} |\hat{r}\rangle + r \dot{\theta} |\hat{\theta}\rangle + \dot{z} |\hat{z}\rangle) \rightarrow$$

$$\ddot{r} |\hat{r}\rangle + \dot{r} d|\hat{r}\rangle/dt + (\dot{r} \dot{\theta} + r \ddot{\theta} |\hat{\theta}\rangle + r \dot{\theta} (-\dot{\theta} |\hat{r}\rangle) + \ddot{z} |\hat{z}\rangle$$

→

$$\ddot{r} |\hat{r}\rangle + \dot{r} \dot{\theta} |\hat{\theta}\rangle + r \ddot{\theta} |\hat{\theta}\rangle - r \dot{\theta}^2 |\hat{r}\rangle + \ddot{z} |\hat{z}\rangle \rightarrow$$

$$|a\rangle = (\ddot{r} - r\dot{\theta}^2) |\hat{r}\rangle + (2\dot{r}\dot{\theta}) |\hat{\theta}\rangle + \ddot{z} |\hat{z}\rangle$$

★ pg 839 exercise 3:

a) $(\nabla \cdot \rho \mathbf{v})$ $\rho = \text{scalar}$

$$\text{div } \rho \langle \mathbf{v} | = \frac{1}{r} \left(\frac{\partial \rho v_r}{\partial r} + \frac{\partial \rho v_\theta}{\partial \theta} + \frac{\partial \rho v_z}{\partial z} \right) \rightarrow$$

$$= \frac{1}{r} \left(\frac{\partial \rho v_r}{\partial r} \right) + \frac{1}{r} \left(\frac{\partial \rho v_\theta}{\partial \theta} \right) + \frac{\partial \rho v_z}{\partial z}$$

b) $[\nabla \cdot \rho \mathbf{v} \mathbf{v}]_r$, ρ is scalar

$$|\text{div } \rho \mathbf{v} \mathbf{v}\rangle \rightarrow = \text{div } \rho |\mathbf{v}\rangle \langle \mathbf{v}| \rightarrow$$

$$\frac{1}{r} \left(\frac{\partial (r \rho |\mathbf{v}\rangle v_r)}{\partial r} \right) + \frac{1}{r} \left(\frac{\partial (\rho |\mathbf{v}\rangle v_\theta)}{\partial \theta} \right) + \frac{\partial (\rho |\mathbf{v}\rangle v_z)}{\partial z} \rightarrow$$

$$\langle \delta r | \text{div } \rho \mathbf{v} \mathbf{v} \rangle = \frac{1}{r} \left(\frac{\partial (r \rho v_r v_r)}{\partial r} \right) + \frac{1}{r} \left(\frac{\partial (\rho v_r v_\theta)}{\partial \theta} \right) \rightarrow$$

$$- \frac{1}{r} (\rho v_\theta v_\theta) + \left(\frac{\partial (\rho v_r v_z)}{\partial z} \right)$$

$$\langle \delta \theta | \text{div } \rho \mathbf{v} \mathbf{v} \rangle = \frac{1}{r} \left(\frac{\partial (r \rho v_\theta v_r)}{\partial r} \right) + \frac{1}{r} \left(\frac{\partial (\rho v_\theta v_\theta)}{\partial \theta} \right) \rightarrow$$

$$+ \frac{1}{r} (\rho v_\theta v_\theta) + \frac{\partial (\rho v_\theta v_z)}{\partial z}$$

$$\langle \delta z | \text{div } \rho \mathbf{v} \mathbf{v} \rangle = \frac{1}{r} \left(\frac{\partial (r \rho v_z v_r)}{\partial r} \right) + \frac{1}{r} \left(\frac{\partial (\rho v_z v_\theta)}{\partial \theta} \right) \rightarrow$$

$$+ \left(\frac{\partial (\rho v_z v_z)}{\partial z} \right)$$

c) $[\nabla \cdot p \mathbf{f}]_\theta \rightarrow |\text{div } p \mathbf{f}\rangle = \text{div } p |\mathbf{f}\rangle \rightarrow = |\nabla p\rangle$

$$\frac{\partial p}{\partial r} \langle \delta r | + \frac{1}{r} \frac{\partial p}{\partial \theta} \langle \delta \theta \rangle + \frac{\partial p}{\partial z} \langle \delta z \rangle \rightarrow$$

$$= \langle \delta \theta | \text{div } p \mathbf{f} \rangle = \frac{1}{r} \left(\frac{\partial p}{\partial \theta} \right)$$

$$d) (\nabla \cdot [\chi \cdot v]) \Rightarrow$$

$$\text{div} \langle v | \chi | = \frac{1}{r} \left(\frac{\partial r \langle v | \chi |}{\partial r} \right) + \frac{1}{r} \left(\frac{\partial \langle v | \chi |}{\partial \theta} \right) + \left(\frac{\partial \langle v | \chi |}{\partial z} \right)$$

$$\langle v | \chi | f_r \rangle = v_r \chi_{rr} + v_\theta \chi_{\theta r} + v_z \chi_{zr}$$

$$\langle v | \chi | f_\theta \rangle = v_r \chi_{r\theta} + v_\theta \chi_{\theta\theta} + v_z \chi_{z\theta}$$

$$\langle v | \chi | f_z \rangle = v_r \chi_{rz} + v_\theta \chi_{\theta z} + v_z \chi_{zz}$$

$$e) [\nabla \cdot \nabla v]_\theta \quad |\nabla v| = \frac{\partial |v|}{\partial r} |f_r| + \frac{1}{r} \left(\frac{\partial |v|}{\partial \theta} \right) |f_\theta| \rightarrow$$

$$+ \frac{\partial |v|}{\partial z} |f_z| \Rightarrow \frac{\partial |f_r|}{\partial \theta} = |f_\theta| \quad \frac{\partial |f_\theta|}{\partial \theta} = -|f_r|$$

$$\frac{\partial |f_z|}{\partial \theta} = 0 \Rightarrow |v\rangle = v_r |f_r\rangle + v_\theta |f_\theta\rangle + v_z |f_z\rangle$$

$$\frac{\partial}{\partial r} |v\rangle = \frac{\partial v_r}{\partial r} |f_r\rangle + \frac{\partial v_\theta}{\partial r} |f_\theta\rangle + \frac{\partial v_z}{\partial r} |f_z\rangle$$

$$\frac{\partial}{\partial \theta} |v\rangle = \left(\frac{\partial v_r}{\partial \theta} - v_\theta \right) |f_r\rangle + \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right) |f_\theta\rangle + \left(\frac{\partial v_z}{\partial \theta} \right) |f_z\rangle$$

$$\frac{\partial}{\partial z} |v\rangle = \frac{\partial v_r}{\partial z} |f_r\rangle + \frac{\partial v_\theta}{\partial z} |f_\theta\rangle + \frac{\partial v_z}{\partial z} |f_z\rangle$$

$$\langle f_\theta | \nabla v | v \rangle = \langle f_\theta | \left(\frac{\partial}{\partial r} |v\rangle \right) v_r + \langle f_\theta | \left(\frac{1}{r} \left(\frac{\partial}{\partial \theta} \right) |v\rangle \right) v_\theta + \langle f_\theta | \left(\frac{\partial}{\partial z} |v\rangle \right) v_z \Rightarrow$$

$$= \left(\frac{\partial v_\theta}{\partial r} \right) v_r + \frac{1}{r} \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right) v_\theta + \left(\frac{\partial v_\theta}{\partial z} \right) v_z$$

$$f) \nabla v + (\nabla v)^t \quad |v\rangle = v_r |f_r\rangle + v_\theta |f_\theta\rangle + v_z |f_z\rangle$$

$$\frac{\partial |f_r\rangle}{\partial \theta} = |f_\theta\rangle; \quad \frac{\partial |f_\theta\rangle}{\partial \theta} = -|f_r\rangle; \quad \frac{\partial |f_z\rangle}{\partial \theta} = 0$$

$$\frac{\partial}{\partial r} |v\rangle = \frac{\partial v_r}{\partial r} |f_r\rangle + \frac{\partial v_\theta}{\partial r} |f_\theta\rangle + \frac{\partial v_z}{\partial r} |f_z\rangle$$

$$\frac{\partial}{\partial \theta} |v\rangle = \left(\frac{\partial v_r}{\partial \theta} - v_\theta \right) |f_r\rangle + \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right) |f_\theta\rangle + \frac{\partial v_z}{\partial \theta} |f_z\rangle$$

$$\frac{\partial}{\partial z} |v\rangle = \left(\frac{\partial v_r}{\partial z} \right) |f_r\rangle + \left(\frac{\partial v_\theta}{\partial z} \right) |f_\theta\rangle + \left(\frac{\partial v_z}{\partial z} \right) |f_z\rangle$$

$$\begin{aligned} |\nabla v| &= \frac{\partial v_r}{\partial r} |f_r\rangle \langle f_r| + \left(\frac{1}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} \right) |f_r\rangle \langle f_\theta| \rightarrow \\ &+ \frac{\partial v_r}{\partial z} |f_r\rangle \langle f_z| + \frac{\partial v_\theta}{\partial r} |f_\theta\rangle \langle f_r| + \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) |f_\theta\rangle \langle f_\theta| \rightarrow \\ &\frac{\partial v_\theta}{\partial z} |f_\theta\rangle \langle f_z| + \frac{\partial v_z}{\partial r} |f_z\rangle \langle f_r| + \frac{1}{r} \frac{\partial v_z}{\partial \theta} |f_z\rangle \langle f_\theta| \rightarrow \\ &+ \frac{\partial v_z}{\partial z} |f_z\rangle \langle f_z| \end{aligned}$$

$$\begin{aligned} |\nabla v|^+ &\rightarrow a |f_j\rangle \langle f_i| \rightarrow \\ &= \frac{\partial v_r}{\partial r} |f_r\rangle \langle f_r| + \left(\frac{1}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} \right) |f_\theta\rangle \langle f_r| \rightarrow \\ &+ \frac{\partial v_r}{\partial z} |f_z\rangle \langle f_r| + \frac{\partial v_\theta}{\partial r} |f_r\rangle \langle f_\theta| + \rightarrow \\ &+ \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) |f_\theta\rangle \langle f_\theta| + \frac{\partial v_\theta}{\partial z} |f_z\rangle \langle f_\theta| \rightarrow \\ &+ \frac{\partial v_z}{\partial r} |f_r\rangle \langle f_z| + \left(\frac{1}{r} \frac{\partial v_z}{\partial \theta} \right) |f_\theta\rangle \langle f_z| + \frac{\partial v_z}{\partial z} |f_z\rangle \langle f_z| \end{aligned}$$

$$|\nabla v| + |\nabla v|^+ \rightarrow = 2 \frac{\partial v_r}{\partial r} |f_r\rangle \langle f_r| + \rightarrow$$

$$\left(\frac{\partial v_\theta}{\partial r} + \frac{1}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} \right) (|f_r\rangle \langle f_\theta| + |f_\theta\rangle \langle f_r|) + \rightarrow$$

$$\begin{aligned}
& + \left(\frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \right) (|f_r\rangle \langle f_z| + |f_z\rangle \langle f_r|) \\
& + 2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) |f_\theta\rangle \langle f_\theta| \rightarrow \\
& + \left(\frac{\partial v_\theta}{\partial z} + \frac{1}{r} \frac{\partial v_z}{\partial \theta} \right) (|f_\theta\rangle \langle f_z| + |f_z\rangle \langle f_\theta|) \rightarrow \\
& + 2 \left(\frac{\partial v_z}{\partial z} |f_z\rangle \langle f_z| \right)
\end{aligned}$$

#2

$$a) \operatorname{div}(p \langle v |) = \langle \nabla p | v \rangle + p \operatorname{div} \langle v |$$

$$\begin{aligned}
\operatorname{div}(p \langle v |) &= \sum_i \frac{\partial p \langle v | f_i \rangle}{\partial q_i} \rightarrow \\
& \underbrace{\sum_i \frac{\partial p}{\partial q_i} \langle v | f_i \rangle}_{\langle \nabla p | v \rangle} + p \underbrace{\sum_i \frac{\partial \langle v | f_i \rangle}{\partial q_i}}_{p \operatorname{div} \langle v |}
\end{aligned}$$

$$b) \operatorname{div} |v\rangle \langle w| = |\nabla v| w \rangle + \langle \operatorname{div} w | v \rangle$$

$$\begin{aligned}
& \downarrow \\
& \sum_i \frac{\partial}{\partial q_i} (|v\rangle \langle w | f_i \rangle) \rightarrow \sum_i \frac{\partial}{\partial q_i} (v_j \langle f_j | \cdot w_i) \rightarrow \\
& = \sum_i \left(\left(\frac{\partial v_j}{\partial q_i} \right) w_i + v_j \left(\frac{\partial w_i}{\partial q_i} \right) \right) |f_j\rangle \rightarrow
\end{aligned}$$

$$= \underbrace{\sum_i \left(\frac{\partial v_j}{\partial q_i} \right) w_i |f_j\rangle}_{|\nabla v\rangle \langle w|} + \underbrace{\sum_i \left(\frac{\partial w_i}{\partial q_i} \right) |v\rangle}_{\langle \text{div } w \rangle}$$

$$c) \text{div} \langle v | \chi | = | \chi | : |\nabla v| + \langle v | \text{div } \chi \rangle$$

$$\downarrow$$

$$\sum_i \frac{\partial}{\partial q_i} \left(\sum_j v_j \chi_{ji} \right) \Rightarrow \sum_{i,j} \left(\frac{\partial v_j}{\partial q_i} \right) \chi_{ji} + v_j \left(\frac{\partial \chi_{ji}}{\partial q_i} \right)$$

$$\sum_{i,j} \frac{\partial v_j}{\partial q_i} \chi_{ji} = | \chi | : |\nabla v|, \quad \sum_{i,j} v_j \left(\frac{\partial \chi_{ji}}{\partial q_i} \right) = \langle v | \text{div } \chi$$